```
    // The content of this payload
    bytes data = 2;
}

// Setting Proposal
// This message proposes a change in a setting value.
message SettingProposal {
    // The setting key.  E.g. sawtooth.consensus.module
    string setting = 1;

    // The setting value. E.g. 'poet'
    string value = 2;

    // allow duplicate proposals with different hashes
    // randomly created by the client
    string nonce = 3;
}

// Setting Vote
// In ballot mode, a proposal must be voted on.  This message
indicates an
// acceptance or rejection of a proposal, where the proposal
is identified
// by its id.
message SettingVote {
    enum Vote {
    ACCEPT = 0;
    REJECT = 1;
    }

    // The id of the proposal, as found in the
    // sawtooth.settings.vote.proposals setting field
    string proposal_id = 1;
    Vote vote = 2;
}
```

Settings transaction family in action

Initially, the transaction processor gets the current values of *sawtooth.settings.vote.authorized_keys* from the state. The public key of the transaction signer is checked against the values in the list of authorised keys. If it is empty, no settings can be proposed, save for the authorised keys.

A *Propose* action is validated. If it fails, it is considered an invalid transaction. A *proposal_id* is calculated by taking the sha256 hash of the raw *SettingProposal* bytes as they exist in the payload. Duplicate *proposal_ids* causes an invalid transaction. The proposal will be recorded in the *SettingProposals* stored in *sawtooth.settings.vote.proposals*, with one 'accept' vote counted. The transaction processor outputs a DEBUG-level logging message similar to the following:

```
"Adding proposal {}: {}".format(proposal_id, repr(proposal_
data)
```

A *Vote* action is validated, checking to see if *proposal_id* exists, and the public key of the transaction has not already voted. The value of *sawtooth.settings.vote.approval_threshold* is read from the state.

- If the 'accept' vote count is equal to or above the approval threshold, the proposal is applied to the state. This results in the above INFO message being logged. The proposal is deleted from the *SettingProposals* record.
- If the 'reject' vote count is equal to or above the approval threshold, then it is deleted from *sawtooth.settings.vote.proposals* and an appropriate debug logging message is logged.

Otherwise, the vote is recorded in the list of *sawtooth.settings.vote.proposals* by the public key and vote pair.

Validation of configuration settings is as follows:

- *sawtooth.settings.vote.approval_threshold* must be a positive integer; it must be between 1 (the default) and the number of authorised keys, and include both.
- *sawtooth.settings.vote.proposals* may not be set by a proposal. END

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